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Final Report: MoCo Real Estate Investing

Introduction

Housing markets have become increasingly important to both investors and policymakers due to rising home prices, affordability concerns, and changing economic conditions. Even within the same county, housing markets can behave very differently depending on local income levels, population growth, and long-term appreciation trends. Montgomery County, Maryland provides a strong case study for housing market analysis because of its diverse cities, strong economic base, and close proximity to Washington, D.C.

This project analyzes housing market trends across Montgomery County using Zillow Home Value Index (ZHVI dataset) data combined with demographic & economic data from the United States Census Bureau. The analysis focuses on identifying long term housing growth patterns, comparing affordability across cities, and evaluating potential real estate investment opportunities through risk adjusted market analysis.

The Zillow dataset provides monthly home value estimates over time, allowing for long-term trend analysis across cities within Montgomery County. Census ACS data was integrated to include variables such as population, median household income, median rent, and median home value. Combining these datasets allowed the analysis to move beyond simple housing price comparisons and evaluate broader economic conditions influencing local housing markets.

ZHVI dataset: <https://www.zillow.com/research/data/>

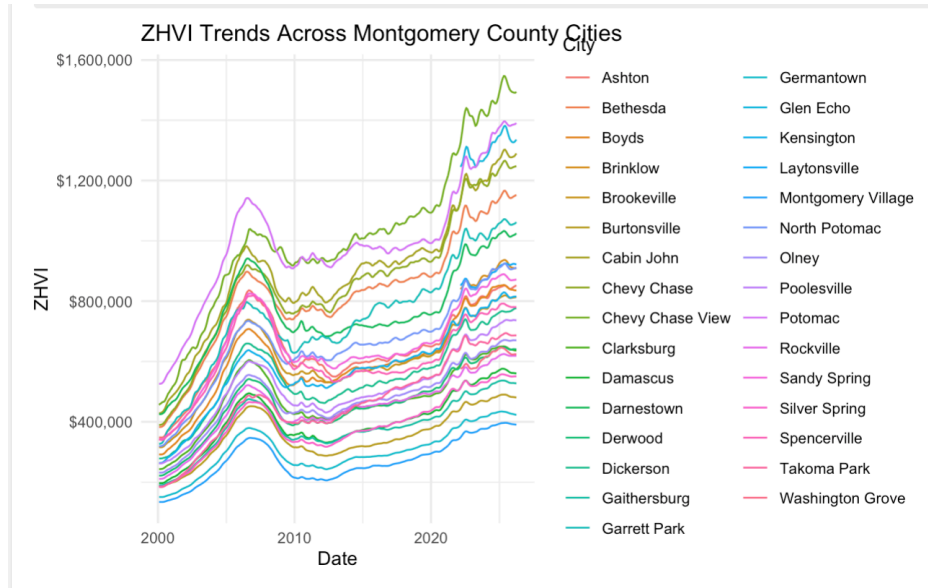
US Census Data: <https://www.census.gov/data.html>

Research Questions

This project was designed to answer several key questions related to housing growth and affordability in Montgomery County:

- How have housing prices changed over time across Montgomery County cities?
- Which cities experienced the strongest long-term housing appreciation?

- Are higher growth housing markets associated with greater price volatility?
- How does affordability differ across cities relative to local income levels?
- Which cities demonstrate the strongest balance between growth, affordability, and investment stability?



Zillow Home Value Index (ZHVI) trends across cities in Montgomery County, Maryland. The visualization highlights long-term housing appreciation patterns and differences in market growth between cities.

Data Cleaning & Pre-Processing

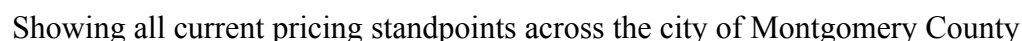
Before beginning the analysis, both the Zillow and Census datasets required cleaning and restructuring to ensure they could work together properly. The Zillow ZHVI data originally stored monthly housing prices in a wide format, so the data was reshaped into a long format in R using `pivot_longer()` to make time series analysis and visualization easier.

The dataset was then filtered to include only cities within Montgomery County, Maryland. Additional cleaning was performed on the date columns to properly format them for analysis and graphing.

One of the biggest challenges involved merging the Zillow and Census datasets because both sources used different naming formats for cities. For example, the Census data included labels

After preprocessing, a final combined dataset was created containing housing prices, population, median income, rent estimates, and other variables used throughout the project's affordability and investment analysis.

Reshaping Data so I could merge the Cencus & Zillow data



Basic Descriptive Statistics

The final dataset once everything was combined included Zillow housing price data & Census economic indicators for cities within Montgomery County, Maryland. The Zillow data provided monthly home value estimates, while the Census data added population, median household income, median rent, & median home value variables.

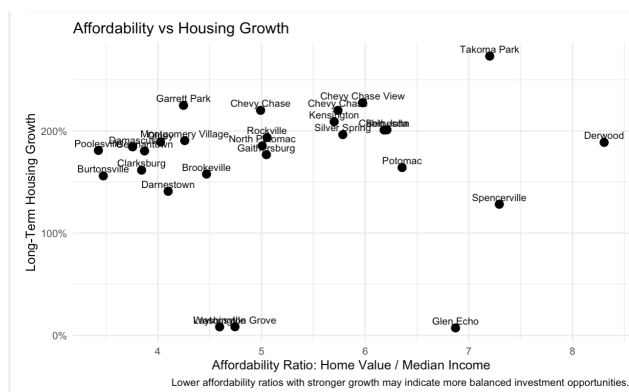
The descriptive statistics showed that housing values varied significantly across Montgomery County cities. Higher cost areas such as Chevy Chase and Garrett Park had some of the strongest home value levels, while cities such as Olney, Poolesville, and Damascus showed more balanced conditions when comparing housing growth, affordability, and stability. The Census variables also showed that income and affordability differed across cities, which helped explain why the most expensive markets were not always the strongest investment opportunities.

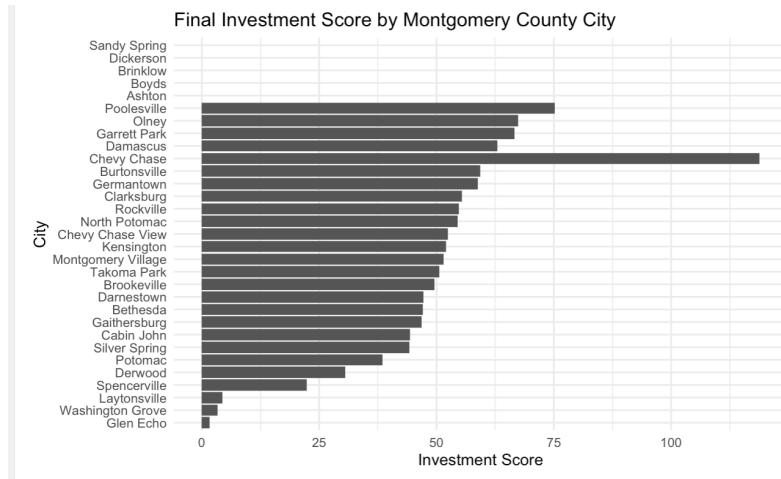
Final Investment Analysis

The final stage of the project focused on combining housing growth, market stability, and affordability into one overall investment analysis. Instead of only looking at rising home prices, the project evaluated which cities showed the strongest long-term balance between appreciation and risk.

An affordability ratio was created by comparing housing prices to median household income levels. A final investment score was then developed using growth, volatility, and affordability metrics together. Cities with strong long-term growth, lower volatility, and better affordability received higher scores.

The results showed that the best investment opportunities were not always the most expensive cities. Instead, some cities demonstrated a stronger balance between growth, stability, and affordability, making them potentially better long-term investment markets.





Conclusion

This project found that Montgomery County housing markets performed very differently across cities. Cities such as Chevy Chase, Poolesville, Garrett Park, Olney, and Damascus ranked highly in the final investment score because they combined strong long-term growth with relatively stable volatility and income support.

The analysis also showed that the “best” investment market is not always the most expensive one. While higher priced areas may show strong home values, cities like Poolesville, Olney, & Damascus demonstrated a better balance between appreciation, affordability, and risk. Takoma Park also showed strong appreciation over time, but affordability conditions need to be considered when evaluating it as an investment market.

Overall, this project shows the value of combining housing data with Census economic indicators. Looking only at price growth can give an incomplete picture, but combining growth, volatility, income, and affordability creates a stronger framework for identifying long-term real estate investment opportunities in Montgomery County.

References

Zillow Data Source: [Zillow Home Value Index \(ZHVI\) Data](#)

Used for monthly home value trends and long-term housing price analysis across Montgomery County cities.

Census API data source: [United States Census Bureau Data](#)

Used for demographic & economic variables including population, median household income, median rent, and median home value.

[American Community Survey \(ACS\) 5-Year Estimates](#)

Used to gather local level economic and demographic data for Montgomery County cities.

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